

Scalability & Future Plan

Date	June 2026
Team ID	XXXXXX
Project Name	Personal Network Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Supports only local deployment	Limits access for multiple concurrent users	High
2	Local JSON storage instead of a database	Limited scalability and data persistence	High
3	Basic authentication and security mechanisms	Not suitable for production deployment	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user	Deploy using cloud services with load balancing and concurrent request handling
2	Data Storage	Local JSON files	Migrate to PostgreSQL or MongoDB with cloud backup
3	Performance	Local model inference	Optimize models, enable GPU inference, and implement response caching
4	Security	Basic validation	Add OAuth2/JWT authentication, HTTPS, encryption, and role-based access control

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment using AWS or Azure	3 Months	Increased availability and scalability
Phase 2	Personalized recommendations using user interaction history	6 Months	Improved user engagement and networking quality

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 3	Integration with LinkedIn and professional networking platforms	9 Months	Enhanced networking opportunities and profile synchronization
Phase 4	Multilingual support and advanced AI models	12 Months	Better accessibility and improved conversation quality